

Field Wave Algebra: A Formal Nonlinear Dynamic Algebra for Generalized Field Theory

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ABSTRACT

Field Wave Algebra (FWA) is formalized as a nonlinear dynamic algebraic framework in which classical linear algebra emerges as a static degenerate limit. The algebra is built upon dynamic operators acting on wave states, deviation operators controlling nonlinear symmetry breaking, and boundary calibration operators governing critical transitions. A rigorous axiom system is introduced, followed by formal definitions of operators, structural properties, and a stability framework based on Lu-calibration. Examples demonstrate the reduction to linear algebra and the stabilization of critical boundaries. FWA is proposed as a mathematically consistent foundation for generalized field theory and dynamic wave-based systems.

1. INTRODUCTION

Classical linear algebra describes static relations between fixed elements. However, many physical systems—such as nonlinear media, photonic architectures, and fractal boundaries—are inherently dynamic. FWA generalizes algebraic structure toward dynamic field behavior.

The goal of this work is to provide a journal-grade formalization of FWA, including:

- axioms of dynamic algebra;
- formal operator definitions;
- deviation and boundary structures;
- nonlinear mode composition;
- reduction to linear algebra;
- Lu-calibration theory;
- examples and applications.

2. PRELIMINARIES

2.1 State Space

Let the space of dynamic wave states be defined as:

$$\Phi = \{ \varphi(x, t) \mid x \in \mathbb{R}^n, t \in \mathbb{R} \}$$

2.2 Operator Classes

We define three fundamental operator classes governing system evolution:

1. Dynamic operator: $F : \Phi \times D \times A \rightarrow \Phi$
2. Deviation operator: $\Delta : \Phi \rightarrow D$
3. Boundary calibration operator: $Lu : B \times \Phi \rightarrow A$

where D is the deviation parameter space, A is the boundary operator space, and B represents boundary surfaces.

3. AXIOMS OF FIELD WAVE ALGEBRA

Axiom 1 (Dynamic Evolution): Every wave state evolves under a dynamic operator:

$$\phi_{t+1} = F(\phi_t, \Delta(\phi_t), A_t)$$

Axiom 2 (Deviation Symmetry Breaking): Deviation operator satisfies:

$$\Delta(\phi) = 0 \iff \phi \text{ is symmetry-preserving}$$

Axiom 3 (Nonlinear Mode Composition): For any two states:

$$\phi = \phi_1 \oplus \phi_2$$

where $\oplus$ is nonlinear and non-distributive.

Axiom 4 (Boundary Interaction): Boundary operator acts as:

$$A_t = Lu(B, \phi_t)$$

ensuring continuity across critical surfaces.

Axiom 5 (Degenerate Static Limit): If $\Delta(\phi) = 0$, $\partial_t \phi = 0$, and $\oplus \rightarrow +$, then:

$$F(\phi, 0, A) \rightarrow A\phi$$

recovering classical linear algebra.

4. FORMAL OPERATOR DEFINITIONS

4.1 Dynamic Operator

A dynamic operator is defined as:

$$F(\phi, \delta, A) = A\phi + N(\phi, \delta)$$

where $A\phi$ is the linear boundary-conditioned part, and $N(\phi, \delta)$ is the nonlinear deviation term.

4.2 Deviation Operator

Deviation is quantified via norm metrics:

$$\Delta(\phi) = \|\partial_t \phi\| + \|\phi \oplus \phi - 2\phi\|$$

4.3 Boundary Calibration Operator (Lu)

$$Lu(B, \phi) = \arg \min_{A \in \mathcal{B}} \|A\phi - \phi\|_B$$

which represents the optimal operator minimizing discontinuity on boundary B .

5. MAIN THEOREMS

Theorem 1 (Reduction to Linear Algebra): If $\Delta(\phi) = 0$ and $N(\phi, 0) = 0$, then:

$$F(\phi, 0, A) = A\phi$$

Proof. Direct substitution into the definition of F .

Theorem 2 (Boundary Stability Under Lu-Calibration): For any boundary B , Lu-calibration ensures:

$$\|A\phi - \phi\|_B \rightarrow 0$$

Proof. Follows directly from the minimization property of the Lu operator, guaranteeing absolute continuity of wave states across critical domains.

6. EXAMPLES

6.1 One-Dimensional Wave

Let $\phi(x, t) = \sin(x - t)$. The deviation evaluates to $\Delta(\phi) = \|\partial_t \phi\| = 1$, and the dynamic evolution follows $\phi_{t+1} = F(\phi, 1, A) = A\phi + N(\phi, 1)$.

6.2 Degenerate Limit

When $\partial_t \phi = 0$ and $N = 0$, we have $\phi(x) = \sin(x)$, yielding $F(\phi, 0, A) = A\phi$, successfully recovering linear algebra.

6.3 Boundary Calibration

Considering a boundary at $x = 0$, the Lu-operator selects A minimizing $\|A\phi - \phi\|_{x=0}$, stabilizing transitions across multi-scale domains.

7. APPLICATIONS

- nonlinear wave propagation systems;
- photonic computing architectures;
- fractal boundary systems;
- generalized field models;
- dynamic operator control systems.

8. CONCLUSION

A complete formalization of Field Wave Algebra has been presented. Dynamic operators, deviation structures, nonlinear composition, and Lu-calibration form a consistent algebraic foundation for generalized field theory. Linear algebra emerges naturally as a static limit, and concrete examples demonstrate the consistency and applicability of the framework.